

COMM114: Basics

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A Continuous Scalar Random Variable

- Let X be a continuous scalar random variable.
- The cumulative density function (CDF) of X is $F_X(a) = P(X \leq a)$
- The probability density function (PDF) of X is $p_X(a) = F'_X(a) = \frac{dF_X(a)}{da}$. $p_X(x)$ satisfies
 - $\int p_X(x)dx = 1$
 - $p_X(x) \geq 0$
- Typical scalar random variables:
 - Gaussian: $X \sim \mathcal{N}(\mu, \sigma)$ where $p_X(x) = \frac{1}{\sigma\sqrt{2\pi}}e^{-(x-\mu)^2/2\sigma^2}$
 - Uniform: $X \sim \mathcal{U}(a, b)$, where $p_X(x) = \frac{1}{b-a}1_{a \leq x \leq b}$

A Continuous Scalar Random Variable (cont.)

- Gaussian: $X \sim \mathcal{N}(\mu, \sigma)$ where $p_X(x) = \frac{1}{\sigma\sqrt{2\pi}}e^{-(x-\mu)^2/2\sigma^2}$
 - mean: $\mathbb{E}[X] = \int xp_X(x)dx = \mu$
 - variance: $\mathbb{E}[(X - \mathbb{E}[X])^2] = \int (x - \mu)^2 p_X(x)dx = \sigma^2$

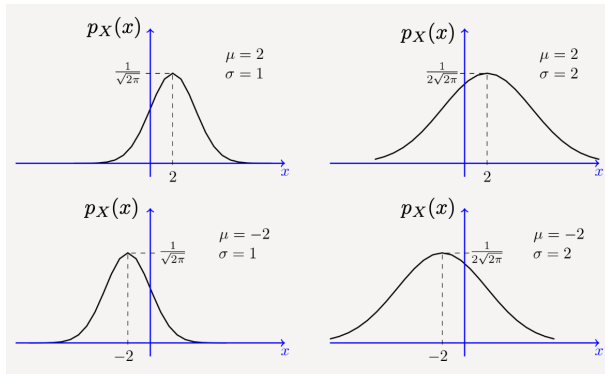


Figure: CDF of Gaussian distribution

A Continuous Scalar Random Variable (cont.)

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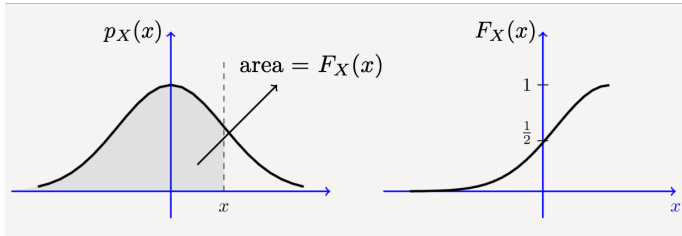


Figure: CDF of Gaussian distribution

A Continuous Scalar Random Variable (cont.)

- Uniform: $X \sim \mathcal{U}(a, b)$, where $p_X(x) = \frac{1}{b-a} 1_{a \leq x \leq b}$

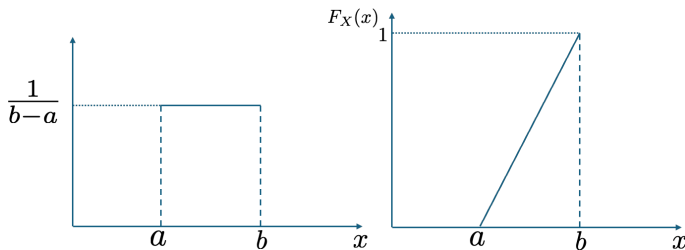


Figure: CDF of Uniform distribution

- mean: $\mathbb{E}[X] = \int x p_X(x) dx = \frac{b+a}{2}$
- variance: $\mathbb{E}[(X - \mathbb{E}[X])^2] = \int [x - (b+a)/2]^2 p_X(x) dx = \frac{(b-a)^2}{12}$

A Discrete Scalar Random Variable

- Let X be a discrete scalar random variable taking values from a discrete set Ω .
- The probability mass function (PMF) satisfies:
 - $P_X(i) = P_X(X = i) \geq 0$ for all $i \in \Omega$
 - $\sum_{i \in \Omega} P_X(i) = 1$
- Typical discrete random variables
 - Bernoulli distribution
 - $\Omega = \{0, 1\}$
 - $P_X(1) = p$ and $P_X(0) = 1 - p$
 - Binomial distribution with parameter n and p
 - $\Omega = \{0, 1, \dots, n\}$
 - $P_X(k) = \binom{n}{k} p^k (1 - p)^{n-k}$

A Continuous multi-dimensional random variable

- Let $\mathbf{X} \in \mathbb{R}^d$ be a continuous multi-dimensional random variable.
- The cumulative density function (CDF) of $\mathbf{X}$ is $F_{\mathbf{X}}(\mathbf{a}) = P(\mathbf{X} \leq \mathbf{a})$, where $\mathbf{a} \in \mathbb{R}^d$
- The probability density function (PDF) of $\mathbf{X}$ is

$$p_{\mathbf{X}}(\mathbf{a}) = \frac{\partial^d F_{\mathbf{X}}(\mathbf{a})}{\partial a_d \cdots \partial a_1} \text{ where } \mathbf{a} = [a_1, a_2, \dots, a_d].$$

$p_{\mathbf{X}}(\mathbf{x})$ satisfies

- $\int p_{\mathbf{X}}(\mathbf{x}) d\mathbf{x} = 1$
- $p_{\mathbf{X}}(\mathbf{x}) \geq 0$
- Typical multi-dimensional random variables:
 - Gaussian: $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ where $p_{\mathbf{X}}(\mathbf{x}) = \frac{1}{|\boldsymbol{\Sigma}|^{1/2} (2\pi)^{d/2}} e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})}$
 - Uniform: $\mathbf{X} \sim \mathcal{U}(\mathbf{a}, \mathbf{b})$, where $p_{\mathbf{X}}(\mathbf{x}) = \left(\prod_{i=1}^d \frac{1}{b_i - a_i} \right) 1_{\mathbf{a} \leq \mathbf{x} \leq \mathbf{b}}$

Chain rule of probability (Continuous case)

A joint probability can be decomposed as a product of conditional probabilities:

- Joint probability of two variables:

$$p(\mathbf{x}, \mathbf{y}) = p(\mathbf{x}|\mathbf{y})p(\mathbf{y}) = p(\mathbf{y}|\mathbf{x})p(\mathbf{x}) \text{ [PDF]}$$

- Joint probability of multiple variables:

$$\begin{aligned} p(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m) &= p(\mathbf{x}_1) \prod_{i=2}^m p(\mathbf{x}_i | \mathbf{x}_1, \dots, \mathbf{x}_{i-1}) \quad [\text{probability density function (PDF)}] \\ &= p(\mathbf{x}_1) \prod_{i=2}^m p(\mathbf{x}_i | \mathbf{x}_{<i}) \end{aligned}$$

Remark: In principle, the index order in the decomposition can be arbitrary.

Chain rule of probability (Discrete case)

A joint probability can be decomposed as a product of conditional probabilities:

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Remark: In principle, the index order in the decomposition can be arbitrary.

Marginal probability

- Continuous case:

$$p(\mathbf{x}_1) = \int_{\mathbf{x}_2} \cdots \int_{\mathbf{x}_m} p(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m) d\mathbf{x}_2 \cdots d\mathbf{x}_m$$

- Discrete case:

$$P(\mathbf{x}_1) = \sum_{\mathbf{x}_2} \cdots \sum_{\mathbf{x}_m} P(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m)$$

Conditional probability

- Continuous case:

$$\begin{aligned} p(\mathbf{x}_m | \mathbf{x}_{<m}) &= \frac{p(\mathbf{x}_m, \mathbf{x}_{<m})}{p(\mathbf{x}_{<m})} \\ &= \frac{p(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m)}{\int_{\mathbf{x}_m} p(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m) d\mathbf{x}_m} \end{aligned}$$

- Discrete case:

$$\begin{aligned} P(\mathbf{x}_m | \mathbf{x}_{<m}) &= \frac{P(\mathbf{x}_m, \mathbf{x}_{<m})}{P(\mathbf{x}_{<m})} \\ &= \frac{P(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m)}{\sum_{\mathbf{x}_m} P(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m)} \end{aligned}$$

Exercise

Suppose the joint distribution of two discrete scalar variables X_1 and X_2 is given by the table below:

$P(x_1 = 1, x_2 = 1) = 0.15$	$P(x_1 = 1, x_2 = 2) = 0.05$	$P(x_1 = 1, x_2 = 3) = 0.1$
$P(x_1 = 2, x_2 = 1) = 0.2$	$P(x_1 = 2, x_2 = 2) = 0.05$	$P(x_1 = 2, x_2 = 3) = 0.05$
$P(x_1 = 3, x_2 = 1) = 0.05$	$P(x_1 = 3, x_2 = 2) = 0.15$	$P(x_1 = 3, x_2 = 3) = 0.2$

- What is $P(x_1 = 1)$?
- What is $P(x_2 = 2)$?
- What is $P(x_1 = 1|x_2 = 1)$?

Bayes' formula

- Continuous case

$$p(\boldsymbol{x}|\boldsymbol{y}) = \frac{p(\boldsymbol{y}, \boldsymbol{x})}{p(\boldsymbol{y})} = \frac{p(\boldsymbol{y}|\boldsymbol{x})p(\boldsymbol{x})}{p(\boldsymbol{y})} \quad [\text{PDF}]$$

- Discrete case

$$P(\boldsymbol{x}|\boldsymbol{y}) = \frac{P(\boldsymbol{y}, \boldsymbol{x})}{P(\boldsymbol{y})} = \frac{P(\boldsymbol{y}|\boldsymbol{x})P(\boldsymbol{x})}{P(\boldsymbol{y})} \quad [\text{PMF}]$$

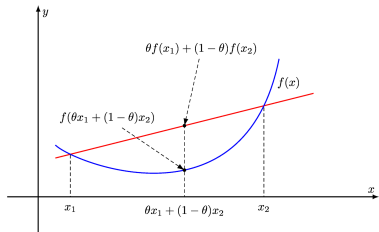
Jensen's Inequality

- Suppose $\mathbf{X}$ is a multi-dimensional random variable and $f(\cdot)$ is a convex function, then we have

$$\mathbb{E}[f(\mathbf{X})] \geq f(\mathbb{E}[\mathbf{X}]).$$

- Conversely, if $f(\cdot)$ is a concave function (i.e., $-f(\cdot)$ is convex), then we have

$$\mathbb{E}[f(\mathbf{X})] \leq f(\mathbb{E}[\mathbf{X}]).$$



Definition of a convex function

Loosely speaking, a function f is convex if for any x_1 and x_2 in the functional domain and $t \in [0, 1]$, we have

$$f(tx_1 + (1 - t)x_2) \leq tf(x_1) + (1 - t)f(x_2)$$

Jensen's Inequality (cont.)

Jensen's Inequality can be slightly generalized into the following form:

- Suppose $\mathbf{X}$ is a multi-dimensional random variable. $f(\cdot)$ is a convex function and $g(\cdot)$ is another function (not necessarily to be convex), then we have

$$\mathbb{E}[f(g(\mathbf{X}))] \geq f(\mathbb{E}[g(\mathbf{X})]).$$

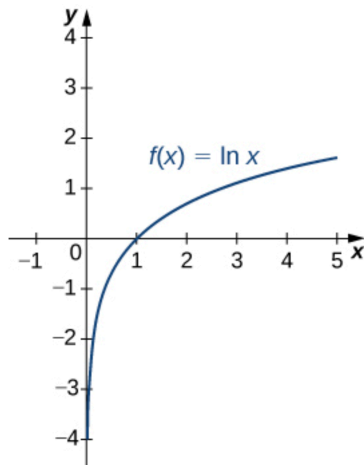
- Conversely, if $f(\cdot)$ is a concave function (i.e., $-f(\cdot)$ is convex), then we have

$$\mathbb{E}[f(g(\mathbf{X}))] \leq f(\mathbb{E}[g(\mathbf{X})]).$$

Jensen's Inequality (cont.)

- Example: The function $\log(\cdot)$ is concave. Therefore, we have

$$\mathbb{E}[\log(g(\mathbf{X}))] \leq \log(\mathbb{E}[g(\mathbf{X})]).$$



Dissimilarity measures of two distributions

- Kullback-Leibler (KL) divergence
- Earth Mover's distance (EMD)

Kullback-Leibler divergence (Continuous case)

The Kullback-Leibler (KL) divergence measures how much a model probability distribution Q is different from a true probability distribution P :

$$D_{KL}(P||Q) = \int p(\mathbf{x}) \log \left(\frac{p(\mathbf{x})}{q(\mathbf{x})} \right) d\mathbf{x}$$

Properties of KL divergence

- Not symmetric.
 - $D_{KL}(P||Q) \neq D_{KL}(Q||P)$
- $D_{KL}(P||Q)$ is always ≥ 0 .
- $D_{KL}(P||Q)$ is equal to 0 if and only if $P = Q$. That is, only if the two probability distribution are identical, $D_{KL}(P||Q) = 0$.

Kullback-Leibler divergence (Continuous case)

Proof of non-negativity of KL divergence

$$\begin{aligned} -D_{KL}(P||Q) &= \int p(\mathbf{x}) \log \left(\frac{q(\mathbf{x})}{p(\mathbf{x})} \right) d\mathbf{x} \\ &\leq \log \left[\int p(\mathbf{x}) \left(\frac{q(\mathbf{x})}{p(\mathbf{x})} \right) d\mathbf{x} \right] \quad [\text{applying } \mathbb{E}[\log(g(X))] \leq \log(\mathbb{E}[g(X)])] \\ &= \log \left[\int q(\mathbf{x}) d\mathbf{x} \right] \\ &= \log 1 \\ &= 0 \end{aligned}$$

Kullback-Leibler divergence (Discrete case)

The Kullback-Leibler (KL) divergence measures the distance between two probability distributions P and Q :

$$D_{KL}(P||Q) = \sum_k P(k) \log \left(\frac{P(k)}{Q(k)} \right)$$

Properties of KL divergence

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Remark: The proof for the discrete case follows the same logic as for the continuous case.

Kullback-Leibler divergence (Discrete case)

KL divergence in terms of entropy and cross-entropy:

$$\begin{aligned} D_{KL}(P||Q) &= \sum_k P(k) \log \left(\frac{P(k)}{Q(k)} \right) \\ &= \sum_k P(k) \log P(k) - \sum_k P(k) \log Q(k) \\ &= - \underbrace{\left[\sum_k P(k) \log P(k) \right]}_{H(P)} + \underbrace{\left[- \sum_k P(k) \log Q(k) \right]}_{H(P,Q)} \end{aligned}$$

- $H(P)$ is the entropy of the probability distribution P .
- $H(P, Q)$ is the cross-entropy of the distribution Q relative to a distribution P .

Applications of cross-entropy

Cross-entropy has been widely used in deep learning:

- Classification tasks with training data $\{\mathbf{x}_i, y_i\}_{i=1}^n$ where y_i is the label for $\mathbf{x}_i$:

$$\begin{aligned} \arg \min_{\boldsymbol{\theta}} \left(- \sum_{i=1}^n \mathcal{L}(\hat{y}_i(\mathbf{x}_i; \boldsymbol{\theta}), y_i) \right) & \quad [\boldsymbol{\theta} \text{ denotes neural network parameters}] \\ = \arg \min_{\boldsymbol{\theta}} \left(- \sum_{i=1}^n \log P_i(\hat{y}_i(\mathbf{x}_i; \boldsymbol{\theta}) = y_i) \right) \end{aligned}$$

Earth Mover's Distance (EMD)

- The earth mover's distance (EMD) measures the dissimilarity between two probabilities P and Q , defined as

$$\text{EMD}(P, Q) = \sup_{\|\nabla f(\mathbf{x})\| \leq 1} \left(\mathbb{E}_{\mathbf{x} \sim P}[f(\mathbf{x})] - \mathbb{E}_{\mathbf{y} \sim Q}[f(\mathbf{y})] \right).$$

- In general, the EMD is computed by searching over possible functions satisfying the constraint $\|\nabla f(\mathbf{x})\| \leq 1$.
- It is difficult for multi-dimensional case.

Earth Mover's Distance (EMD) (cont.)

- 1-dimensional continuous case:

$$\text{EMD}(P, Q) = \int |F_1(x) - F_2(x)| dx,$$

where $F_1(x)$ is the CDF function for P and $F_2(x)$ is the CDF function for Q .

Questions?